MOSS Core V0.7 - ready-terminated harness

Prototype quotation, 1 November 2026. Quote 2 kits, plus separate prices at 10 / 50 / 100 kits.

One labelled and electrically tested kit serves one rover. Separate tooling/setup, test fixture, sourcing, lead time and shipping to Thailand. Price the documented STOP switch with its wired lid tail separately, and optional device adapters only where existing leads do not fit.

Supply H02 and nine-wire H03 form one sleeved Cytron cable ensemble with TWO Core terminations: J11 for supply and J2 for controls plus four winding outputs. H03 has no supply-positive conductor. J2.5 is signal ground; driver supply current uses J11. Leave J2.10 empty. Neither winding lead is ground.

CAD lengths follow the relaxed connector wire-exit to wire-exit paths, including the service slack drawn. Prototype quote free span = longest modelled conductor + 20 mm, rounded up to 5 mm; target tolerance +/-5 mm. The assembler adds cut/strip, contact engagement and termination allowances. These are quotation estimates, not released cut lengths. Fit the first article before freezing the batch. Complete USB cables are selected commercially and are not cut to CAD length.

Route	Function	Wires / cable	CAD longest mm	Quote span mm
H01	Battery supply	2	163.7	185
H02	Cytron supply	2	121.9	145
H03	Cytron combined link	9	237.3	260
H04	Left motor and encoder	6	55.8	80
H05	Right motor and encoder	6	128.7	150
H08_chassis	STOP / chassis half	4	128.6	150
H08_lid	STOP / removable lid tail	4	113.6	135
O01	Arm bus input	3	242.4	265
O02	Speaker lead	2	66.0	90
O03_left	Fan 1 / left cooling fan	2	207.6	230
O03_right	Fan 2 / right cooling fan	2	132.5	155
U01	Orange Pi POWER	Complete cable	90.7	Select cable
U02	Core USB data	Complete cable	136.9	Select cable
U03	Hub upstream / Pi DATA	Complete cable	207.8	Select cable

14 routes = 42 custom wire segments, including the optional adapters, + 3 complete USB cable assemblies. H03's centreline is the common trunk; individual wire lengths include fan-out. Left/right refer to forward driving. Final cut/strip lengths follow first-article fitting.

Cytron and motor pin schedule

J2 = nine populated circuits in a ten-way housing. J3/J4 = complete six-wire motor cables.

Core	Function / net	Cytron endpoint	Wire / Core contact
J2.1	MDD_PWM1	Cytron PWM1	24 AWG / 43030-0001
J2.2	MDD_DIR1	Cytron DIR1	24 AWG / 43030-0001
J2.3	MDD_PWM2	Cytron PWM2	24 AWG / 43030-0001
J2.4	MDD_DIR2	Cytron DIR2	24 AWG / 43030-0001
J2.5	GND	Cytron control GND	24 AWG / 43030-0001
J2.6	CYTRON_L_LEAD	Cytron M1A	18 AWG / 43030-0038
J2.7	MOTOR_L_RETURN	Cytron M1B	18 AWG / 43030-0038
J2.8	CYTRON_R_LEAD	Cytron M2A	18 AWG / 43030-0038
J2.9	MOTOR_R_RETURN	Cytron M2B	18 AWG / 43030-0038
J2.10	UNUSED	No wire or contact	Empty cavity

J3 left / J4 right	Function	Retained motor pigtail	Wire / Core contact
1	M1	M1 winding	18 AWG / 43030-0038
2	M2	M2 winding	18 AWG / 43030-0038
3	ENC_GND	Encoder GND	24 AWG / 43030-0001
4	ENC_3V3	Encoder VCC, switched 3.3 V	24 AWG / 43030-0001
5	ENC_A	C1 / channel A	24 AWG / 43030-0001
6	ENC_B	C2 / channel B	24 AWG / 43030-0001

Motor loops: M1A -> J2.6 -> left shunt -> J3.1, and M1B -> J2.7 -> J3.2. Right: M2A -> J2.8 -> right shunt -> J4.1, and M2B -> J2.9 -> J4.2. Both winding legs are H-bridge outputs; neither is ground.

Retain both six-wire pigtails supplied with Arduitronics RU00129. With power disconnected, trace each stock wire to the motor board's M1/M2, GND/VCC and C1/C2 markings. Record cavity numbers, wire gauge and insulation OD, then approve the factory wirelist. Device-end cavity numbering and OEM connector MPN are not established. Quote insulated factory transitions when the stock wire cannot be terminated directly with the specified Core contact. Never force 18 AWG into an unverified stock motor contact.

Power, STOP and device connections

Read the numbered cavities on each exact connector; display colours do not establish a pinout.

Route / Core	Numbered connection	Termination / note
H01 / J10	1 negative / GND; 2 fused battery positive	Existing fused 3S battery lead; retain external 15 A fuse; XT30 cable extension only if needed
H02 / J11	1 switched battery +; 2 GND	Cytron POWER + / -; one sized ferrule per Core cavity
O01 / J5	1 DATA; 2 switched battery +; 3 GND	Arm servo daisy chain; optional adapter; one common power path
O02 / J7	1 speaker +; 2 speaker -	Both amplifier terminals driven; neither is GND
O03 / J14, J15	1 fused 3S positive; 2 GND	Two nominal 12 V fans; up to 12.6 V; shared F3 1 A fuse
U01	Core J20 -> Orange Pi POWER	Complete USB-C to USB-C 5 V / 3 A cable
U02	HB1 downstream -> Core J1	Complete USB-A to USB-C data cable
U03	Orange Pi DATA -> OTG -> HB1 upstream	Retain HB1 captive 800 mm USB-A cable; store excess clear of moving parts

Core J8	X08 circuit	Switch terminal	Function
1	1	C1	CUT source
2	2	NC1	CUT return
5	3	C2	STATUS source
6	4	NC2	STATUS return
3 / 4	None	None	No contacts or wires

STOP: quote Yijia YJ-LA16-22ZS/A, red, non-illuminated, two independent NC loops. Leave NO terminals unused and insulated. An existing unmarked button requires an unpowered continuity check; six tabs alone do not prove the needed contacts.

X08 chassis 51021-0400 + 4 x 50058-8000; lid 51047-0400 + 4 x 50133-8000. 28 AWG, insulation OD 0.50-0.80 mm. Manufacturer circuits 1-4 pass straight through. The pair uses friction retention; support the service loop and disconnect by the housings. Only H08_lid moves with the cover.

Connector parts and wire selection

Installed quantities per rover; PCB headers are already in the PCBA BOM.

Connection	Qty	Selected cable-side reference	Specification
J2	1	Molex 43025-1000	10-way housing; fit cavities 1-9; cavity 10 empty
J3/J4	2	Molex 43025-0600	6-way housings; six contacts each
Micro-Fit power	8	Molex 43030-0038	18 AWG: 4 in J2 + 2 in each motor housing
Micro-Fit signals	13	Molex 43030-0001	24 AWG: 5 in J2 + 4 in each motor housing
J8	1 + 4	SHR-06V-S + SSH-003T-P0.2-H	Six-position JST SH housing; contacts 1/2/5/6 only
X08 chassis	1 + 4	51021-0400 + 50058-8000	PicoBlade female; 28 AWG, OD 0.50-0.80 mm
X08 lid	1 + 4	51047-0400 + 50133-8000	PicoBlade male; same circuit numbers 1-4
Motor device ends	2	Supplied Arduitronics RU00129 pigtails	Retain original six-way plugs; trace and record functions
Cytron device end	1 set	Supplied 5-way control lead + screw terminations	Match printed PWM1/DIR1/PWM2/DIR2/GND and M1A/B/M2A/B
J11 supply	2	Ferrules sized to selected supply wire	Existing Phoenix 1715721 PCB block; one wire per cavity
J10 extension	If needed	AMASS XT30U-F / XT30U-M	Retain external 15 A pack fuse; omit extension if existing lead reaches
STOP button	Separate option	Yijia YJ-LA16-22ZS/A	Red, non-illuminated, two independent NC contacts
Arm adapter O01	Optional	50-37-5033 + 3 x 08-70-1040	Mates vertical Molex 22-03-5035; retain existing arm daisy chain
Speaker O02	Optional	PHR-2 + 2 x SPH-002T-P0.5S	Mates vertical JST B2B-PH-K-S(LF)(SN)
Fans O03	Optional, two	XHP-2 + 2 x SXH-001T-P0.6 each	Mates JST B2B-XH-A(LF)(SN); verify actual fan polarity

Use stranded copper and document the wire manufacturer, gauge and insulation OD. For the selected Micro-Fit hand-tool processes: 18 AWG OD 1.60-1.85 mm and 24 AWG OD 1.42-1.85 mm. The terminal product's maximum OD is 1.85 mm. The battery/Cytron supply's 16 AWG entry is a quotation candidate retained from the existing supply concept; qualify actual supply leads separately.

Visual routing: winding wire OD 1.8 mm, control/encoder OD 1.5 mm; sleeve envelopes about 7.1 mm OD for H03's common trunk, 5.3 mm for its four-output power branch and 6.1 mm for each complete motor cable. Supplier dimensions remain to be confirmed on the prototype.

Workmanship and first article

The quotation is ready for supplier review; physical fitting and electrical qualification are still pending.

Retain both six-wire pigtails supplied with Arduitronics RU00129. With power disconnected, trace each stock wire to the motor board's M1/M2, GND/VCC and C1/C2 markings. Record cavity numbers, wire gauge and insulation OD, then approve the factory wirelist. Device-end cavity numbering and OEM connector MPN are not established. Quote insulated factory transitions when the stock wire cannot be terminated directly with the specified Core contact. Never force 18 AWG into an unverified stock motor contact.

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Provide a numbered mating-face drawing and factory wirelist, crimp tooling, wire specification and finished-length tolerance. Label each assembly at both ends. Identify LEFT/RIGHT motor cables and both Core terminations of the single Cytron loom. Quote and document all insulated factory splices/transitions.

Test every kit for continuity, shorts and polarity. With power disconnected, both STOP loops must be closed when released and open when pressed or X08 unplugged. Check isolation between the two loops. Keep J2.10 and J8.3/4 empty.

Fit the first kit with cover closed and opened. Inspect connector seating, wire exit/bends, access to the disconnect, strain relief, fan clearance, sharp edges and storage of the HB1's excess captive lead. Freeze batch spans and cut/strip allowances after fitting.

Planned encoder test: compare counts and direction to an independent rotation reference during forward/reverse motion and repeated loaded reversals. Check for missed/extra counts, resets and supply dips. Check the complete wire/crimp assembly under startup/load for temperature and voltage drop; the Cytron channel rating is not the harness rating.

A factory-terminated kit avoids maker crimping. For a small DIY quantity, 43030-0007 is Molex's bagged loose-piece alternative to reel 43030-0001 for 20-24 AWG. Use 63828-0200 for the listed 18 AWG / 43030-0038 combination and 63819-0000 for its listed 20-30 AWG contacts. The 18 AWG tool specifies optimal insulation OD 1.60-1.85 mm; the 20-24 AWG tool scope specifies 1.42-1.85 mm. Record the exact wire and measure insulation OD; AWG alone does not establish crimp compatibility. Thick silicone wire may not fit. Pre-crimped leads are another documented option, but verify their exact terminal, gauge, insulation and length; no specific pre-crimp lead MPN is selected here.

Evidence: the independent static connector check matched 76 pins on 12 connectors. It does not establish physical motor-end cavity mapping, crimp quality or powered operation. Cable-housing geometry is an explicitly dimensioned visual reference because exact manufacturer STEP/IGES was not obtained. Use current viewer outputs; the previous film is historical.

Sources and revision identity

Manufacturer sources for contacts/tooling; distributor SKU for the retained stock motor kit.

S1 - Molex 43025-0600, six-way cable housing

S2 - Molex 43025-1000, ten-way cable housing

S3 - Molex 43030-0038, 18 AWG female contact

S4 - Molex 43030-0001, 20-24 AWG female contact, reel

S5 - Molex 43030-0007, loose-piece packaging alternative

S6 - Molex 63828-0200 hand-tool scope and crimp data

S7 - Molex 63819-0000 hand-tool scope and crimp data

S8 - Molex pre-crimped leads overview

S9 - Arduitronics RU00129 motor kit and supplied cable

S10 - Cytron MDD10A official product/manual

S11 - Yijia LA16 catalogue: two independent changeover contacts

S12 - Molex PicoBlade 51047/51021 mating drawing

S13 - JST SH, six-position STOP housing/contact

S14 - JST PH, speaker housing/contact

S15 - JST XH, fan housings/contacts

S16 - Molex 5264 arm housing drawing

S17 - Molex 5263 arm contact drawing

S18 - Hoco HB1: four USB-A sockets and 80 cm captive lead

S19 - Molex Micro-Fit current derating, populated circuits

S20 - Phoenix 1715721 supply terminal block

S21 - AMASS XT30U cable connectors

Full URLs are supplied in SOURCES.md and HARNESS_SPECIFICATION.json.

PCB SHA256: 7d88c146a2482b4d89a26b8dbafe18132ae649e58c58671ddaed233b6ab8bc14

ROUTE_LENGTHS.json SHA256: ffa1e1283457c61096f7128c7c20471511b5268852f9a02089c626a8f6d9c91f